

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

HireMind AI

AI-Powered Applicant Tracking and Resume Analysis Platform

1. Introduction

1.1 Purpose

The purpose of HireMind AI is to streamline the recruitment process by automating resume screening, candidate tracking, ATS scoring, and interview preparation using Artificial Intelligence. The system helps recruiters efficiently manage job applications while assisting candidates in improving their resumes.

1.2 Scope

HireMind AI is a web-based platform developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js). The system provides functionalities such as resume analysis, ATS score generation, job management, candidate tracking, AI-powered recommendations, and interview question generation.

1.3 Objectives

- Automate resume screening.
 - Generate ATS compatibility scores.
 - Track applicants through different hiring stages.
 - Improve recruiter productivity.
 - Provide resume improvement suggestions.
 - Generate AI-based interview questions.
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2. Overall Description

2.1 Product Perspective

HireMind AI is an independent recruitment management system that integrates AI services to analyze resumes and assist recruiters in hiring decisions.

2.2 Product Functions

Candidate Module

- User Registration/Login
- Profile Management
- Resume Upload
- Resume Analysis
- ATS Score Generation

- Resume Improvement Suggestions
- Job Search and Application
- Interview Preparation

Recruiter Module

- Recruiter Registration/Login
- Company Profile Management
- Job Posting
- Candidate Search
- Application Management
- Candidate Status Tracking
- Interview Scheduling

Admin Module

- User Management
- Recruiter Management
- Job Monitoring
- Analytics Dashboard
- Report Generation
- System Monitoring

AI Module

- Resume Parsing
 - Skill Extraction
 - ATS Score Calculation
 - Resume Suggestions
 - Interview Question Generation
 - Candidate Matching
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2.3 User Classes

Candidate

- Creates profile
- Uploads resume
- Applies for jobs
- Views ATS score

Recruiter

- Posts jobs
- Reviews applications
- Tracks candidates
- Generates interview questions

Admin

- Manages platform

- Monitors activities
 - Generates reports
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2.4 Operating Environment

Frontend

- React.js
- HTML5
- CSS3
- JavaScript
- Bootstrap/Tailwind CSS

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Integration

- OpenAI API / Gemini API

Development Tools

- Postman
 - IntelliJ
 - Git & GitHub
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3. Functional Requirements

FR-1 User Authentication

Description

Users must register and login securely.

Inputs

- Name
- Email
- Password

Outputs

- Authentication Token
 - User Dashboard
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FR-2 Resume Upload

Description

Candidates can upload resumes.

Inputs

- PDF/DOCX Resume

Outputs

- Resume stored in database
 - Resume analysis initiated
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FR-3 Resume Analysis

Description

System extracts candidate information.

Extracted Data

- Name
- Skills
- Education
- Experience
- Certifications
- Projects

Outputs

- Structured candidate profile
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FR-4 ATS Score Generation

Description

System evaluates resume quality.

Factors

- Keyword Match
- Skill Relevance
- Experience
- Formatting
- Education

Outputs

- ATS Score (0-100)
 - Improvement Suggestions
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FR-5 Job Management

Description

Recruiters can create and manage jobs.

Features

- Create Job
 - Edit Job
 - Delete Job
 - View Applicants
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FR-6 Job Application

Description

Candidates apply for jobs.

Outputs

- Application Record
 - Recruiter Notification
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FR-7 Candidate Tracking

Stages

- Applied
- Screening
- Shortlisted
- Interview Scheduled
- Selected
- Rejected

FR-8 AI Interview Question Generator

Description

Generate interview questions based on:

- Job Role
- Skills
- Experience

Outputs

- Technical Questions
 - HR Questions
 - Coding Questions
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FR-9 Dashboard and Analytics

Candidate Dashboard

- ATS Score
- Applications
- Resume Insights

Recruiter Dashboard

- Total Jobs
- Total Applications
- Hiring Statistics

Admin Dashboard

- User Statistics
 - Recruiter Statistics
 - Platform Analytics
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4. Non-Functional Requirements

Performance

- Response time below 3 seconds.
- Support 1000+ concurrent users.

Security

- JWT Authentication

- Password Encryption using bcrypt
- HTTPS Communication
- Role-Based Access Control

Reliability

- 99% uptime.
- Automatic error handling.

Scalability

- Cloud deployment support.
- Horizontal scaling support.

Availability

- Accessible 24×7.

Usability

- User-friendly interface.
 - Responsive design for mobile and desktop.
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5. Database Requirements

Users Collection

- userId
- name
- email
- password
- role

Resumes Collection

- resumeId
- userId
- filePath
- atsScore
- analysisData

Jobs Collection

- jobId
- recruiterId
- title
- description
- skillsRequired

Applications Collection

- applicationId
- candidateId
- jobId
- status

Interviews Collection

- interviewId
 - applicationId
 - scheduledDate
 - questions
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6. System Architecture

Presentation Layer

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React Frontend

Application Layer

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Node.js + Express.js API

Business Logic Layer

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Authentication

Resume Analysis

ATS Engine

Job Management

Data Layer

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MongoDB Database

External Services

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OpenAI / Gemini API

7. Future Enhancements

- Video Interview Analysis
- AI Resume Builder
- Real-Time Chatbot
- LinkedIn Integration
- Email Automation

- AI Career Guidance
 - Multi-Language Support
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8. Conclusion

ResumeIQ – Smart Resume Analysis & ATS Scoring that combines Applicant Tracking System (ATS) functionalities with Artificial Intelligence to improve hiring efficiency. The system benefits recruiters by reducing manual screening efforts and helps candidates optimize their resumes for better job opportunities.